Element B student work

Thursday, September 25, 2025 12:56 PM

Bilal Qureshi
Environmental, Behavioral, or enrichment
Bella – Red Eyed Slider
Zark – Western Painted Shell

https://awionline.org/lab-animal-search/lane-s-williams-w-mayo-m-2022-impact-environmental-enrichment-red-eared-slider

- What Worked
  - Turtles with more rocks and plants showed more natural behaviors, foraging and sheltering
  - Negative behaviors decreased (less stress, less abnormal movement)
  - Showes Enrichment directly improves Well-being and activity
- What didn't work
  - Not all behaviors improve (basking didn't change a lot)
  - Some enrichment may not affect all behaviors equally

Things I found include how the turtles use more natural resources to support their natural behavior and survival. I also think that some types of natural resources can be hard to maintain, such as real plants. Also, having some rocks that can be too sharp for the turtles. Another aspect is providing only one shelter for one of the turtles, which can cause behavioral decline if the turtle is in the shelter for a long time, leaving the other turtle in distress. This can also cause the turtles to refrain from the time they bask at the basking area per day, hence weakening the health of the turtles.

https://pmc.ncbi.nlm.nih.gov/articles/PMC8833317/

- What worked
  - Enrichment improves
    - Physical + mental health
    - Natural behavior
    - Survival after release
- Helps turtles adapt back to the wild
- What didn't work
  - Mostly used in rehabilitation centers, not small habitats
  - Requires planning and recourses

I found that using resources or making the habitat similar to the real world can help prepare the turtles for the release after they are healthy and able to thrive on their own in the wild. Natural behaviors are adapted by the environment they are in. Building something for their Behavioral environment takes a long time and can take a lot of trial and error to make them comfortable with change. This also takes a lot of resecures due to the trial and error.

https://www.sciencedirect.com/science/article/pii/S0168159125002497

- What worked
  - Animals in enriched environments
    - Were more active
    - Showed less stress
    - Preferred natural habitats
  - Confirms animals choose enrichment when given the option
- What didn't work
  - Study done on lizards, not turtles
  - May not apply to all turtle species

This study found that reptiles in enriched environments were more active, showed reduced stress, and preferred naturalistic habitats over simple ones. These findings support the idea that natural habitat features improve animal well-being. However, the study focused on reptiles broadly and not specifically on turtles, which may limit how directly the results apply. This suggests that while naturalistic design is important, it must be adapted to meet the specific needs of turtle species.

https://marinesavers.com/2022/07/turtles-environmental-enrichment/

- What worked
  - Encouraged exploration and curiosity
  - Improved fitness and natural instincts
- What didn't work
  - Different turtles respond differently
  - Requires trial and error

This project demonstrated that enrichment activities, such as introducing objects and varying environmental conditions, increased turtles' curiosity, physical activity, and overall fitness. This shows that interactive and changing environments can successfully improve turtle behavior. However, the effectiveness of enrichment varied between individual turtles, meaning that some strategies may not work universally. This highlights the need for flexible and adaptable habitat designs.

https://awionline.org/lab-animal-search/diggins-r-burrie-r-ariel-e-et-al-2022-review-welfare-indicators-sea-turtles

- What worked
    - Identified Effective Enrichment types
        - Feeding enrichment
        - Structural enrichment
    - Behavior is a good measure of success
- What didn't work
    - Some enrichment methods don't work well
    - Limited research for specifically for turtles

This review identified several types of enrichment, including feeding and structural enrichment, that are effective in improving turtle welfare. It also emphasized that observing behavior is key to determining success. However, the study noted that not all enrichment methods are effective for turtles and that research in this area is still limited. This indicates that careful selection of enrichment strategies is necessary, as poorly chosen methods may not improve habitat quality.

Note from Mr Martz
WELL DONE on finding detailed, credible work to support your work. This is excellent!